

INTEGRATING SOCIAL CONTEXT INTO EMOBERT-
GCN FOR SENTIMENT ANALYSIS OF
FILIPINO POLITICAL DISCOURSE

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The thesis hereto attached entitled INTEGRATING SOCIAL CONTEXT INTO EMOBERTGCN FOR SENTIMENT ANALYSIS OF FILIPINO POLITICAL DISCOURSE, prepared and submitted by GABRIEL LUIGI L. VIRTUCIO, in partial fulfillment of the requirements for the degree of BACHELOR OF SCIENCE (Computer Science) is hereby accepted.

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ABSTRACT

GABRIEL LUIGI L. VIRTUCIO, University of the Philippines Los Baños, MAY 2026.
**INTEGRATING SOCIAL CONTEXT INTO EMOBERTGCN FOR SENTIMENT
ANALYSIS OF FILIPINO POLITICAL DISCOURSE**

Major Professor: ASST. PROF. ARIEL B. DORIA

With the rise of political tensions both internationally and locally in the Philippines, the analysis of online political discourse becomes a source of crucial information to derive public opinion towards government policies, with elections serving as the primary context for sentiment extraction. However, political events often carry polarizing opinion which create imbalanced datasets, which remains one of the major hurdles in sentiment analysis. This study aims to address that concern by extending the BertGCN model by injecting conversational context into the graph structure of its GCN component, enabling sentiment to propagate across reply chains grounded in Emotional Contagion theory and the principle of Homophily. Using a corpus of 73,799 tweets from the 2025 Philippine general midterm elections, the proposed model was evaluated against a RoBERTa baseline fine-tuned for code-switched Tagalog sentiment and the original EmoBertGCN. Ten-fold cross-validation shows macro F1 scores of 85.3% (RoBERTa baseline), 84.6% (BertGCN baseline), and 84.9% (extended BertGCN), indicating that the proposed integration yields no significant performance gains ($p > 0.05$ for both comparisons). Despite the null finding, this research contributes a manually and machine-annotated dataset of 73,799 election-period tweets as a resource for future Tagalog NLP research. A BERT-LDA topic modelling approach was then utilized to yield distinct topic clusters for deeper sentiment analysis. The study contributes new datasets in Filipino as well as provides methodological insights for future graph-based sentiment analysis in low-resource, conversation-sparse environments.

INTRODUCTION

Natural Language Processing (NLP) has evolved from the rule-based and keyword-driven systems into sophisticated models capable of understanding the tone, context, and emotion in text. Among the many categories of NLP, sentiment analysis has been pivotal in understanding and gauging human sentiment towards specific topics/subjects. This specific branch of NLP provides researchers, organizations, companies, and governments with meaningful information that could improve their services, projects, or processes. In recent years, the sentiment classification task has been powered by deep learning and transformer models, which include all the Bidirectional Encoder Representations from Transformers (BERT) variants (Devlin et al., 2018). These transformer models have achieved significantly improved accuracies and are now deemed state-of-the-art models. However, these new models are still limited by the amount of corpus there is for the specific language under training. Languages that lack large and well-annotated datasets are called low-resource languages, which these new models have difficulty understanding.

At the same time, with the rapid evolution of technology and digital platforms, more and more activities and processes have now become easier to accomplish. In the context of the political landscape, government officials and key political figures can now easily reach their audience and even members of other political groups. Social media has now become a primary avenue for political discourse. This phenomenon can be seen all over the globe, and in some regions, such as in Southeast Asia, social media now plays a transformative but complex role in the politics of the countries in that region (Takahashi, 2025). Other studies local to the Philippines have also shown the dangers of unchecked use of digital platforms towards the outcome of elections, both regional and national (Takahashi, 2025; Yusingco, 2021). In addition to that, Filipinos have also been shown to spend almost four hours a day on social media, the most of any country in the world (Dixon, 2025). One of the unintended consequences of increased social media usage rate is the rise in political polarization among

the population (Batac & Cordel, 2023).

Despite the effectiveness of transformer-based models, recent literature has pushed for new frameworks to improve sentiment classification. Some of these approaches now include the entire corpus when classifying sentiment. They now employ the use of Graph-based learning, specifically Graph Neural Networks (GNNs), to take advantage of the entire corpus when classifying individual documents. Foundational models such as BertGCN (Lin et al., 2021), preceded by TextGCN (Yao et al., 2018), combine graph-structured learning with transformer embeddings to model semantic relationships across large corpora. More recent work by Aquino et al. (2025) improves upon BertGCN by preserving emojis, hashtags, and mentions in the preprocessing stage, retaining even more context for the GNN component to learn upon. However, despite these advancements, these new frameworks still lack the conversational context that social media posts contain.

This research seeks to extend the EmoBertGCN framework by including social interaction edges, specifically conversation-based relationships. In doing so, the sentiment analysis framework gains even more context. Using the 2025 Philippine general midterm elections as the corpus to train on, the research also provides valuable resources and insights for future NLP research in the context of the Philippines. As a result, the study then provides not only a methodological improvement upon previous iterations of the BertGCN, but also a deeper understanding of the political expression of Filipinos regarding local issues in online spaces.

Statement of the Problem

Despite advancements in transformer-based sentiment analysis through the procurement of more quality Tagalog corpora, pre-training of more language models, and integration of Graph Neural Networks in sentiment classification, several critical challenges still remain:

1. Sentiment imbalance in political datasets restricts skew model training and leads to biased and unreliable sentiment prediction.
2. Current GNN-integrated sentiment models (BertGCN) capture document-word and word-word relationships, but ignore social relationships between posts. Consequently, the framework fails to take advantage of emotional propagation and the interactional context within social media conversations.
3. There is limited research applying these newly pre-trained language models and advanced architectures to Philippine-specific datasets, especially ones situated during electoral periods, which would provide significant political and social insight towards public sentiment.

Research Objectives

The objectives of this research can be broken down into the following goals. The first objective is determine if the inclusion of conversational context in sentiment analysis of political discourse would improve model performance in low-resource and class-imbalanced datasets. The second objective is to create more resources for the application of NLP in Tagalog Corpora through the creation of a labeled resource on the Philippine 2025 general midterm elections for further research to better capture and understand the context during that key political event.

Significance of the Study

This research is significant in both NLP tasks and socio-political analysis. In the natural language processing domain, this research advances sentiment analysis for low-resource languages by extending the EmoBertGCN framework to incorporate social interaction edges. In doing so, this framework addresses both the lack of data and pretraining for Tagalog- and

Filipino-code-mixed text and the incomplete representation of relationships among social media posts. In the socio-political domain, this paper provides a foundation for examining online political discourse in the Philippines. Consequently, this research is beneficial to computational linguists, social scientists, and policymakers concerned with dynamic public sentiment in online spaces. In relation to the United Nations Sustainable Development Goals (SDGs), this study addresses two of the goals. First is SDG 9 (Industry, Innovation, and Infrastructure), as the study aims to strengthen existing sentiment analysis methods to capture fully and accurately the public political sentiment of Filipinos. This also relates to SDG 16 (Peace, Justice, and Strong Institutions), as the results of this study would better equip political leaders to improve and adapt their governance to actual public sentiment.

Theoretical Framework

This study is founded upon two theoretical frameworks which was utilized to enhance sentiment analysis over online political discourse. Emotional Contagion theory suggests that individuals' emotional states are influenced by and can influence other individuals (Coviello et al., 2014; Hatfield et al., 1993). The principle of Homophily postulates that individuals who share similar characteristics (demographics, behaviors, beliefs, etc.) are likely to interact with one another (Halberstam & Knight, 2016; McPherson et al., 2001). Together, these theories provide theoretical justification that online interactions carry meaningful information for deep learning models to learn from. Methodologically, this translates to incorporating conversation-based relationships (specifically posts under the same conversation thread) when constructing the graph for the model to learn from and train on.

Scope and Limitations

This study focuses on sentiment analysis of Twitter (X) data from the 2025 Philippine general midterm elections. The scope includes text in Tagalog, English, or code-mixed formats. As a result, the results may not generalize to other social media platforms or key political events. The methodology only covers the integration of social interaction edges into EmoBertGCN. It also includes evaluation using quantitative performance metrics and topic modelling to interpret model inference. The implementation of social edges is limited to only conversation-based relationships. This means that tweets are only connected if and only if one is in reply to another. This study does not consider the temporal attributes of the posts, interaction metrics, or the unique users associated with individual posts in the dataset. This decision was made to isolate the impact of conversational structure itself, independent of user identity or timing, a direction that can be explored in future work. The anonymization of unique user identifiers is done to maintain data privacy.

Definition of Terms

The following terms are defined conceptually and operationally as they are extensively used in this study.

Sentiment Analysis. A task or process that involves using AI-driven frameworks in order to analyze, identify, and categorize emotions or sentiments within texts. This is most commonly used on data such as reviews, social media posts, etc. In this study, the task was focused on the analysis of sentiment within online political discourse pertaining to the 2025 Philippine Midterm Elections.

Class imbalance. A major issue present in most, if not all, Natural Language Processing (NLP) tasks where target category distributions are skewed (Henning et al., 2023). This study also faces a class balance issue, in which the sentiment labels are skewed towards negative and neutral labels.

Low-resource languages. In NLP, languages that are less studied, resource scarce, less computerized, less privileged, less commonly taught, or low-density are identified as low-resource (Cieri et al., 2016). The Filipino language, which includes all the different dialects under it, is classified as such. Consequently, this is also another issue that this study faces.

Transformer models. This is a type of deep learning neural network architecture that processes sequential data in parallel rather than sequentially (Stryker & Bergmann, 2023; Vaswani et al., 2017). This model served as one of the primary architectures for the hybrid model explored in this study.

Bidirectional Encoder Representations from Transformers (BERT). This is a transformer-based neural network model made primarily for NLP. Its key feature is its bidirectional processing of texts, which allows it to understand its meaning based on the context of its surrounding text (Devlin et al., 2018). In this study, BERT is used in tandem with a Graph Neural Network (GNN).

Robustly Optimized BERT Pretraining Approach (RoBERTa). This is an optimized variant of BERT. It introduces the following pretraining objectives: dynamic masking, sentence packing, larger batches, and a byte-level BPE tokenizer (Liu et al., 2019). This study utilized models pretrained in the Tagalog language, which use the RoBERTa architecture.

Graph Neural Networks (GNN). A class of neural networks specifically designed to train and perform inference over a graph input. Sanchez-Lengeling et al. (2021) describe it as an optimizable transformation on all attributes of the graph (nodes, edges, global-context) that preserves graph symmetries. In this study, it is the class of one of the component models in the BertGCN framework.

EmoBertGCN. A hybrid model framework that utilizes existing pretrained BERT models and a Graph Neural Network. The model is an extension of the BertGCN but now incorporates emojis, hashtags, and mentions – present in social media posts – to improve

sentiment classification performance (Aquino et al., 2025). In this study, EmoBertGCN was explored further with the inclusion of social media post interactions, specifically conversation threads.

Emotional Contagion. The theory that emotions and emotional state can be transferred from person to person (Hatfield et al., 1993). This also applies to online interactions (Coviello et al., 2014). In this study, it is posited that interactions between different users under posts contain emotional contagion, which, if included in the graph input structure for the GNN, will provide additional information for the EmoBertGCN to learn from.

Homophily. The principle that in a network, connections are likely to exist in nodes sharing similar characteristics. In the context of human interaction, this idea posits that people’s personal networks are homogeneous with regard to many sociodemographic, behavioral, and intrapersonal characteristics (McPherson et al., 2001). This study used this as a foundation for the incorporation of conversation-based edges in the graph input structure.

Heterophily. The opposite of homophily in which connections form when nodes have dissimilar characteristics. In the social setting, this is defined as the degree to which pairs of individuals who interact are different in certain attributes, such as beliefs, education, or socio-economic status (Rogers, 1983). In this study, this is identified as a possible risk towards sentiment analysis performance (Zhu et al., 2020); nevertheless, it is argued that this is a necessary trade-off for a more realistic analysis of political sentiment in social media.

Conversation-based relationships (Social edges). The core innovation of this study is the inclusion of the conversation context in the creation of the graph input for the EmoBertGCN hybrid model framework. Specifically, this refers to posts that are in the same conversation thread.

REVIEW OF LITERATURE

This research is founded upon the evolution of sentiment analysis, the introduction of graph-based learning in sentiment analysis, and the strengths and limitations of existing graph-based text classification frameworks. The understanding and utilization of all three contribute to further improving sentiment analysis in the Philippines context.

Sentiment Analysis

Sentiment analysis has progressed from rudimentary methods to highly sophisticated and complex context-aware systems. Early approaches classified text using sentiment lexicons or word lists; however, these methods lacked the ability to capture nuance, sarcasm, or idiomatic expressions. Sentiment lexicons are predefined lists of words assigned positive, negative, or neutral sentiment scores. The sentiment analysis framework then evolved to utilize traditional machine learning and deep learning techniques, such as Long Short-Term Memory (LSTM), which is a type of recurrent neural network (RNN) capable of learning order dependence in sequence prediction, and Convolutional Neural Network (CNN), a neural network primarily used for processing data with grid-like topology, which had the ability to capture and classify sentiment with high accuracy, where the previous iterations had difficulty with (Jiang et al., 2021).

Currently, leading sentiment analysis models are transformer-based, like the Bidirectional Encoder Representations from Transformers (BERT) and the Multilingual Bidirectional Encoder Representations from Transformers (mBERT) (Devlin et al., 2018). These models are essentially a stack of transformer encode layers (Vaswani et al., 2017). The novelty of BERT compared to models such as LSTM and CNN is its ability to read backwards and forwards simultaneously to acquire complex and insightful word and phrase representations. During its pre-training stage, it incorporates the use of Masked Language

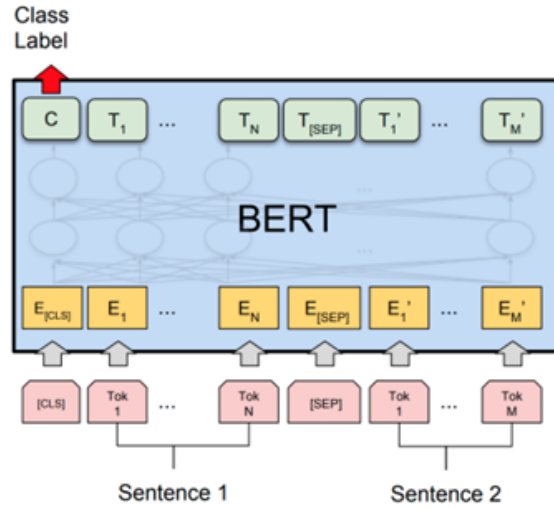


Figure 1. BERT Fine-tuning (Devlin et al., 2018)

Modeling (MLM), in predicting randomly masked words, and Next Sentence Prediction (NSP), in understanding sentence relationships. Afterwards, the pre-trained model can now be further fine-tuned to handle more specific tasks, one of them being text classification (in the context of this research, sentiment analysis). In sentiment analysis, the document input is first trained into a pre-trained BERT, then the classification [CLS] token is extracted and fed into the output layer for classification. Figure 1 illustrates the fine-tuning process of BERT-based models for tasks such as sentiment analysis.

The year after the publication of BERT, Liu et al. (2019) found that BERT was significantly undertrained and, if addressed, could match or exceed the performance of every model published after it. Hence, they introduce a new model called Robustly Optimized BERT Pretraining Approach (RoBERTa), which improves upon BERT on several factors, such as longer training with more data, larger batch sizes, removal of NSP Loss, the use of dynamic masking, longer sequences, and the use of Byte-Level Byte-Pair-Encoding. The removal of NSP loss was due to ablation studies, such as the paper by Yang et al. (2020), which question its importance. RoBERTa's usage of Byte-Level Byte-Pair-Encoding further breaks down the previously character-level into bytes to encode any input text without using an "unknown" token, essentially capturing tokens previously ignored by

Table 1. Evolution of Pretrained Language Models for Philippine Languages

Aspect / Paper	Pretraining Corpus	Models Pretrained & Released	Benchmark Datasets Introduced/Used
(Cruz & Cheng, 2019)	WikiText-TL-39 (~39M tokens)	BERT (Cased/Uncased), AWD-LSTM (for ULMFiT)	Private Sentiment Dataset
(Cruz & Cheng, 2020)		BERT (Cased/Uncased, WWM), DistilBERT (from BERT Cased)	Hate Speech (Binary), Dengue (Multilabel)
(Jiang et al., 2021)	Multi-source Corpus (~1.3GB, ~270M tokens from OSCAR, Wiki, News) TLUnified (Massive corpus from Bitexts, OSCAR, NewsPH) New Corpus (Formal & Informal texts from websites/social media; ~8.2M lines for Tagalog, ~2M for Cebuano)	BERT (Uncased), ELECTRA (Uncased), RoBERTa Base	Hate Speech (Binary), Dengue (Multilabel), NewsPH-NLI, POS Tagging
(Cruz & Cheng, 2021)		RoBERTa Base, RoBERTa Large	Hatespeech (Binary), Dengue (Multilabel), NewsPH-NLI
(Visperas et al., 2023)		BERT (Tagalog & Cebuano), RoBERTa (Tagalog & Cebuano)	Sentiment Analysis, POS Tagging, NER datasets

BERT (emojis, typographical errors, etc.). All these key features lead to a model that achieves state-of-the-art results.

In the context of sentiment analysis of Filipino content, mBERT is able to perform reasonably well, given its 104-language training enables zero-shot transfer between English and Tagalog within the same sentence. In a study conducted by Maceda et al. (2023), they demonstrated mBERT’s promising performance with an accuracy and F1 score of 80.21% and 81.14% with only minimal pre-processing when handling code-mixed text (Tagalog and English). While mBERT provides a solid baseline, it still struggles with languages that are identified as low-density, less commonly taught, under-resourced, and low-resource in nature, which Tagalog falls under (Cieri et al., 2016; Singh, 2008).

The evolution of these models for Philippine languages is documented in Table 1. Early work by Cruz and Cheng (2019) established a baseline with the WikiText-TL-39 corpus (~39M tokens) and BERT-based models. Afterwards, Jiang et al. (2021) introduces a corpus of approximately 270 million tokens by incorporating OSCAR and news data. However, the most substantial leap came from Cruz and Cheng (2021) with the TLUnified corpus, which, combined with the RoBERTa architecture, set new benchmarks for hate speech detection and natural language inference. Recently, Visperas et al. (2023) extended

this by pretraining BERT and RoBERTa models for Cebuano, using large corpora of formal and informal texts. Most importantly, they introduced a model specifically trained for sentiment analysis (positive, negative, neutral). This illustrates a shift towards larger, more diverse, and informally-sourced corpora (including social media text) and the utilization of the RoBERTa architecture to better understand the context of low-resource languages like Tagalog and Cebuano.

Despite the gains made possible by the introduction of new pre-trained models and corpus texts, a fundamental issue remains which is class imbalance, which happens when the distribution of class labels favors one or more majority classes (Henning et al., 2023). This issue becomes even more obvious in political datasets. According to Schöne et al. (2021), tweets with negative emotional language were considerably more likely to be retweeted (shared) than those with positive language. Positive language, on the other hand, was a much weaker predictor than negative language even in the positive contexts. Consequently, the minority class, which in this case is the positive sentiment class, is systematically underrepresented, leading to lower per-class performance on those labels even when overall accuracy appears acceptable (Sebastiani, 2015).

Some common solutions to class imbalance include resampling, data augmentation, and modifying the loss function (Henning et al., 2023). While effective in certain settings, these strategies address the frequency of minority-class samples without utilizing the social media structure surrounding those samples. A minority-class document, regardless of how many times it is resampled, still carries only its own local context when processed by a sequential model such as BERT. The model has no mechanism to draw on the broader distributional patterns of the corpus when classifying any individual document.

Graph-based frameworks such as TextGCN and BertGCN offer a structurally different solution to this problem. By representing the entire corpus as a heterogeneous graph and performing node classification through message passing, these models allow individual documents to aggregate information from connected nodes through edges based on each

document’s lexical features. This means that even a minority-class document benefits from the collective signal present in its graph neighborhood, rather than being classified purely in isolation. Yao et al. (2018) and Lin et al. (2021) both note that their respective frameworks achieve strong performance even under limited training data conditions, which is attributable in part to this corpus-wide information sharing. In this sense, graph-based learning offers a representational solution to class imbalance that complements, rather than replaces, data-level mitigation strategies.

Graph-Based Text Classification Methods

In an effort to improve sentiment analysis, frameworks that include Graph Neural Networks (GNNs) have been introduced. A comprehensive review conducted by Wang et al. (2024) states that while numerous recent text classification models applied the sequential deep learning technique (methods that process text information in order, such as RNN, LSTM), graph neural network-based models can directly deal with complex structured text data and exploit global information that may be overlooked by sequential models.

Formally introduced by Scarselli et al. (2009), GNNs were proposed to tackle the limitations of traditional graph-based algorithms and better represent non-Euclidean relations in practical applications. They are neural network models configured to represent data about entities and their relationships (Noble, 2025). Another key feature of GNNs is its ability to model diffusion of information, as they have a mechanism called message passing, where information is repeatedly aggregated to central nodes from their neighborhood (Han et al., 2025). Some notable and state-of-the-art models include but are not limited to Graph Convolutional Networks (GCN), Graph Attention Networks (GAT), and GraphSAGE (Hamilton et al., 2017; Kipf & Welling, 2016; Veličković et al., 2017).

A notable example of GNN being used for text classification is in the study entitled “Graph Convolutional Networks for Text Classification” (Yao et al., 2018), which was proposed with the intention of designing a model trained on the entirety of the corpus

represented as a heterogeneous graph where words and documents are represented as nodes. In their construction of the graph input, they defined the edges and weights in two ways:

- (a) **Document–Word edges** are word occurrences in documents and weighted by Term Frequency–Inverse Document Frequency (TF-IDF) to reflect the importance of a term across documents.
- (b) **Word–Word edges** are word co-occurrence information in the corpus weighted by Pointwise Mutual Information (PMI), which is used to capture semantic co-occurrence within a sliding window.

This framework enables global word co-occurrence information beyond word sequences, which are often overlooked by sequence-based models such as BERT and CNN. This graph input is then fed into a two-layer GCN (Kipf & Welling, 2016), which transforms the text classification problem into a node classification problem. GCN jointly learns embeddings for both words and documents, which is supervised by the known class labels for documents, transforming text classification into a node classification task. The framework achieves state-of-the-art results on several benchmarks without relying on pre-trained word embeddings, demonstrating robustness when faced with limited training data. In the context of sentiment analysis, this innovation provides a framework for modeling non-sequential relationships between words and documents.

TextGCN is later built upon by BertGCN, proposed by Lin et al. (2021). They proposed a hybrid model that combines the large-scale pretraining of BERT models with the transductive learning of GCN. Extending upon TextGCN, BertGCN constructs the heterogeneous graph where the document nodes are initialized using BERT embeddings rather than an identity matrix. Other characteristics of the graph structure, such as the word-document edges and the word-word edges (which now use a Positive Pointwise Mutual Information (PPMI)), are mirrored in this new proposed hybrid framework. Furthermore, they employ the following new features: (a) the use of a memory bank mechanism to

ensure efficiency when handling large-scale datasets, and (b) an interpolated loss function that combines predictions from both BERT and GCN outputs, which forces the model to balance both information sources. In summary, the initialized document embeddings, which originate from BERT, are refined in the GCN component, which is used to compute the final prediction. Afterwards, during the backpropagation, BERT learns to classify text that returns optimal results from both the direct classification of BERT and the graph-based refinement of GCN, essentially using the context of the corpus through term importance and term co-occurrence. BertGCN achieves state-of-the-art results on several text classification benchmarks, especially on longer documents where the GCN component provides significant gains.

A more recent and relevant study in the Philippine context by Aquino et al. (2025) directly addresses the methodological gaps in sentiment analysis for low-resource languages, specifically Tagalog. The authors proposed the EmoBertGCN hybrid model, which employs the use of the previously proposed BertGCN but now incorporates emojis, hashtags, and user mentions to improve semantic understanding on online sentiment. In this approach, they treated these tokens as essential features for semantic and sentiment understanding rather than noise to be filtered out during the preprocessing stage. They employed the use of their model on an imbalanced dataset of Filipino political tweets in which they achieved a macro F1-score of 68.72%, outperforming a wide range of traditional machine learning, deep learning, and transformer-based baseline models. Additionally, while LLMs like GPT-4o achieved higher metrics, the authors emphasized the cost and computation efficiency of EmoBertGCN. Furthermore, they extended their text classification task by introducing a topic modeling framework that combines BERT embeddings with Latent Dirichlet Allocation (LDA) and Log-Likelihood Ratio (LLR) to yield more distinct topic clusters for deeper sentiment analysis. However, now that BertGCN and EmoBertGCN demonstrate state-of-the-art performances, a remaining challenge is the implementation of document-to-document relationships, especially in the context of applying these hybrid

sentiment analysis frameworks on social media.

Emotional Contagion

This study builds upon the discussed Bert and EmoBertGCN hybrid models using the idea of emotion propagating in social networks. In doing so, the hybrid model is provided with more context to work with when learning the relationships between not only documents and words and words to words, but also between documents. Emotional contagion is defined by the transfer of emotional states from person to person, leading individuals to experience the same emotion as the people around them (Hatfield et al., 1993). Additionally, the premise that emotions can spread through social networks was established by a seminal work by Coviello et al. (2014), where they conducted an experiment to see whether an individual whose emotion is rain-induced can be used to predict a change in emotion of their friends. Findings show that both positive and negative emotions are contagious online, specifically that (a) positive posts are more contagious than negative ones, (b) positive and negative emotions inhibit the spread of each other, and (c) the total indirect effect of rainfall on emotional expression is 150% more than its direct effect, demonstrating that emotional expression in social media is not generated in isolation, but is influenced by the emotional states of one's social connections. Another study by Kramer et al. (2014) experimentally proves this phenomenon, though this study was later critiqued for its manipulation of users' feeds to induce a change in emotion without their express approval (McNeal, 2014). Despite this, the study's findings provide proof of emotional contagion in social networks.

Assumption of Homophily

Building upon the proof of emotional contagion in online social platforms, studies have also shown that similar thinking individuals tend to group themselves together. A landmark paper titled "Birds of a Feather: Homophily in Social Networks" describes social

networks as being strongly shaped by shared characteristics (McPherson et al., 2001). This pattern remains true even in online political discourse (specifically in Twitter, now known as X), as the study by Halberstam and Knight (2016) establishes that homophily and group size of similar-thinking users systematically bias the flow of political information on social media. Furthermore, this concept of homophily has also been used to describe GNNs as a prevalent characteristic, often implied, in many foundational GNN architectures, which is their assumption of homophilous graph structures. This can be seen with the use of aggregation schemes, which work well when connected nodes share similar features and labels (Zhu et al., 2020).

Despite the dominant homophily characteristic of social networks online, it is also important to note that, though homophily dominates social networks, heterophily may still exist. Studies like that of De Francisci Morales et al. (2021) show that cross-cutting interactions happen even when politically segregated communities exist. Though De Francisci Morales et al.’s study is situated in a different media platform (Reddit), this demonstrates the existence of these types of interactions. Given this understanding of the social network structure, it is also important to discuss the clear struggle that GNNs face when handling heterophilous graphs. Empirical benchmarks reveal that heterophilic graphs remain a challenge for default advanced models. Zhu et al. (2020) demonstrates that even Graph Attention Networks (GAT) can struggle on graphs with strong heterophily, as their attention mechanisms may fail to reliably identify and leverage connections to dissimilar nodes. The same can be said for GraphSAGE, as though it performs better compared to GAT and GCN by a large margin, it still underperforms against a simple MLP that ignores the graph structure entirely. These results further demonstrate the point that heterophily remains a fundamental challenge in state-of-the-art GNN models.

Research Gap

While BertGCN and EmoBertGCN successfully integrate document-word and word-word relationships within a heterogeneous graph structure, they share the same limitation of ignoring the relationship between documents. In this framework, document nodes are only conditionally independent given shared lexical features (word nodes). This limitation forces the model to only learn from semantic co-occurrence patterns and is not equipped to propagate contextual information between document nodes.

This limitation is especially apparent in datasets obtained from social media, where posts are inherently relational within interaction structures such as comments, replies, reposts and the like. In political discourse, these replies contain vital information regarding the user's agreement, disagreement or emotional reaction to the parent post. Existing literature does not incorporate these interactional structures into the corpus-level graph. Hence, this research extends BertGCN and EmoBertGCN by incorporating social edges, representing conversation thread relationships, to create pathways through which emotional contagion can propagate.

Furthermore, the inclusion of these conversation-thread relationships is grounded in the previously discussed theories of emotional contagion, which states that the emotional states can transfer between users via exposure (Coviello et al., 2014; Kramer et al., 2014). While these studies establish contagion through passive feed exposure, this study posits that direct engagement can also act as a channel for emotional influence. Unlike indirect engagements, posts belonging to a conversational chain signifies a focused investment in a specific topic, making it a solid pathway for sentiment to flow within a conversation thread. Architecturally, this extension makes use of the message-passing mechanism of the GCN. A reply-chain of Posts A, B and C allows the representation of Post A to be aggregated into the representation of Post B and C, and vice versa. Consequently, the model will learn to leverage the connection between posts to better classify them.

This architectural choice also provides a direct mechanism to mitigate the class imbalance problem noted earlier. In politically charged social media datasets, minority sentiment classes are by definition underrepresented in the training distribution, which can lead a graph-based model to propagate majority class signals more strongly simply due to their higher frequency. By introducing conversation-based edges, a post belonging to a minority class can receive aggregated information from its parent or child posts along conversation thread pathways. Emotional contagion theory thus serves a dual role: it justifies why sentiment can flow along conversation edges, and it explains how underrepresented posts might benefit from neighbor representations during GCN message passing — especially when their own features are insufficiently discriminative. Without these edges, each post’s classification relies primarily on its own lexical content and local word co-occurrence patterns. With them, emotional influence propagates across conversation threads, enabling the model to potentially borrow predictive signal from connected posts. Whether this borrowing improves or degrades performance in the presence of heterophilous interactions (e.g., disagreement edges encoding opposite sentiments) is precisely the question this research aims to discover.

By integrating conversation-based edges, the framework aims to construct a graph that faithfully represents the interactional dynamics present within the social media environment. The integration of these conversation edges when constructing the corpus-level graph will enable the BertGCN to realistically capture the sentiment using conversational context by providing a pathway for the GCN to learn relationships between posts. However, it is important to note that the addition of the conversation-based edges may encode both homophilous and heterophilous interactions as established in other social media platforms like Reddit (De Francisci Morales et al., 2021). Therefore, the integration of these conversation edges when constructing the corpus-level graph will enable the BertGCN to realistically capture political discourse sentiment.

METHODOLOGY

Data Preparation

This stage converts raw social media data into an attributed, heterogeneous graph suitable for GNN training. The dataset was sourced from a user named BwandoWando (2025). Tweets were collected from December 24, 2024 up until May 31, 2025 in the context of the 2025 Philippine midterm elections with the use of a 3rd party service named twitterapi.io. The following keywords were used in scraping the tweets:

1. *Bobotante, Botante, Botohan, Bumoto, Comelec, Commission on Elections, Eleksyon, Eleksyon2025, Halalan, Halalan2025, Iboto, Kandidato, Philippine Elections, Philippine Politics, Senado, Senador.*

The collection was done every 12:00am Philippine Standard Time (PST, UTC+8) and only captures tweets created within the past 24 hours. This process is repeated daily. The following data were collected: Tweet Content, Engagement Metrics (such as Retweet count, Reply count, Like count, Quote count, View count, Bookmark count), Date posted, Author Username, Username of the person being replied to, Conversation ID. All tagged usernames (ex: @username) and forms of IDs are replaced with a unique hashid value based on original value retaining data integrity.

Given that the midterm elections overlapped with an ongoing local event called the “Pinoy Big Brother: Celebrity Collab Edition” which had an occasional audience participation via voting, a significant number of these tweets were collected. To address these issues, the data was filtered by these following characteristics:

1. The dataset was filtered to tweets that either contained or is in a conversation thread with a post that contained the following keywords:

- (a) *Comelec, Commission on Elections, Eleksyon, Eleksyon2025, Halalan, Halalan2025, Kandidato, Philippine Elections, Philippine Politics, Senado, Senador*

2. PBB-related content, specifically that containing these key words, such as core PBB show terms, contestants, ship/duo nicknames, and fan culture terms.

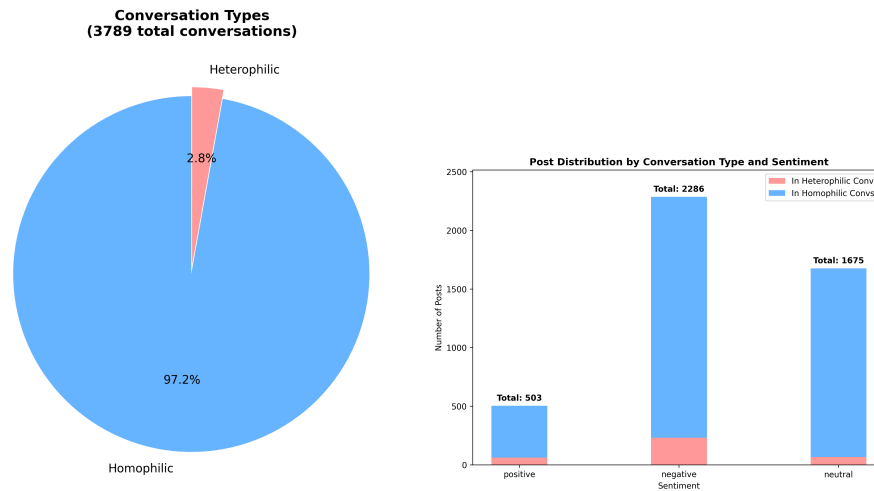
Additionally, the dataset was filtered to only include posts that contained Tagalog content. This was done using the text classification tool fastText (Joulin et al., 2016) without additional fine-tuning to accurately identify and incorporate only tweets that include Tagalog text. After filtering the corpus, the dataset size was reduced from 217,640 to 123,864 tweets. The preprocessing used in Cruz and Cheng (2021)'s paper and in Aquino et al. (2025)'s paper was used. These steps are as follows:

1. HTML meta text and link texts are replaced with a special token [LINK], preserving the presence of a link without keeping the noise of a URL.
2. Mentions within the text are retained (tokens preceded by an @, e.g. @user).
3. Any token preceded by a # are retained (tokens preceded by an #, e.g. #Halalan2025).
4. Punctuation and apostrophe's are normalized to avoid odd token splits.
5. Non-ASCII characters, special symbols, and extra white spaces were removed.
6. Truncation indicators were removed.
7. Emojis are replaced with meaningful text form (e.g., a smiling face emoji becomes "smiling face").
8. Any leading or trailing quotation marks and extra spaces are stripped from the text.

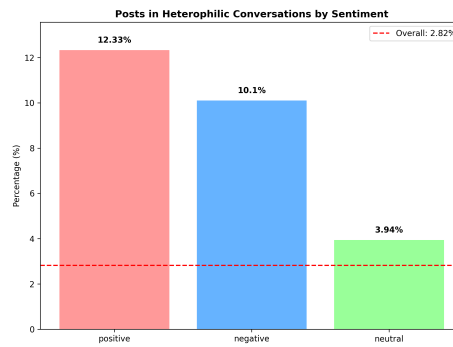
During the manual labeling stage, the annotation framework from Maceda et al. (2023) was adopted. Here, tweets was classified into three categories: neutral, negative, and positive

which was encoded as numbers 2, 1, and 0 respectively. Positive sentiments are defined as expressions of satisfaction, happiness, admiration, interest, and gratitude. Negative sentiments are defined as expressions of dissatisfaction, anger, disappointment, sarcasm, mockery, and frustration. Lastly, Neutral sentiments are defined as no discernible sentiment, conveying information or facts. A questionnaire, adopted from Mohammad (2016), was given out to selected native speakers to guide their annotation as shown in Table 2. The guidelines provided to the annotators were validated by an expert in the field to ensure annotation quality. In the selection of tweets to be annotated, a stratified random sample of approximately 5% of the total tweets was drawn from the preprocessed dataset to ensure representation across different time periods. During the annotation stage, each annotator was provided with an overlap of 120 posts which was given out in intervals over 2 weeks. The Cohen’s Kappa statistic for the labeling process was calculated to be 0.77, indicating that the results of the model’s inference could still be used for tentative conclusions (Carletta, 1996; Krippendorff, 2019).

Prior to the experiment, an examination of the dataset’s graph structure was conducted. It was revealed that the dataset features sparse conversation connectivity in the training set. Only 1,051 of 4,464 posts (23.5%) participated in conversation threads with two or more posts. The remaining 3,413 posts (76.5%) existed as isolated nodes with no conversation edges. The 376 conversation threads consisted of about 2.8 posts each, indicating mostly shallow interaction chains. However, among the few posts that are connected, most of them are homophilic in nature. As illustrated by the subfigures in Figure 2, approximately 97.2% of the subset of connected tweets are homophilic in nature. The negative sentiment class constitutes a majority of the connected posts, with around 10% of negative posts being connected to other posts with differing sentiment labels. Although the positive class has the highest percentage of heterophilic connected posts, its actual number of connected posts is significantly lower than that of the negative and neutral classes, with only 503 connected posts. These findings indicate that the risk of heterophilic posts would be insignificant



(a) Percentage of posts under homophily vs. heterophily (b) Distribution of connected posts under each sentiment label.



(c) Percentage of posts under heterophily in each sentiment label.

Figure 2. Homophily and Heterophily analysis of dataset

Table 2. Annotation Guidelines for Sentiment Classification

Label	Sentiment	Description
0	POSITIVE	The speaker is using positive language toward a candidate, platform, or election-related topic. Examples: support, admiration, enthusiasm, forgiveness, fostering success.
1	NEGATIVE	The speaker is using negative language toward a candidate, platform, or election-related topic. Examples: criticism, pessimism, questioning validity/competence, failure, sarcasm, ridicule or mockery.
2	NEUTRAL	The speaker does not use discernible sentiment or conveys information/facts about the 2025 elections. May contain balanced positive/negative sentiments about the primary subject.

towards the training of the GCN component of the hybrid model. However, the sparsity of the connected posts posed a new risk of insignificant returns from integrating conversation edges between posts.

Given that the dataset was sourced from a publicly available repository and conversation connectivity was an inherent characteristic of the collected data rather than a controllable variable, augmenting or replacing the dataset was not feasible within the time constraints of this study. As such, the experiment proceeded under sparse conditions, which itself reflects a realistic deployment scenario where conversation data may be naturally limited.

Data Training and Inference

Following Aquino et al. (2025), this study extends the **EmoBertGCN** architecture by incorporating **social interaction edges** into the graph structure, as shown in Figure 3. The original model integrates document-word (TF-IDF) and word-word (PPMI) relations. This study extended the EmoBertGCN model by adding a third edge type: document-to-document edges, defined by reply relationship. Heterogeneous graph $G = (V, E)$ was constructed, where Equation. 1 describes the nodes and Equation. 2 describes the edges to be implemented.

The feature of a document node is defined in Equation. 3 where the document embedding was obtained from a BERT model, while the feature of the word node is initialized to zero. The weight of the edges are further defined in Equation. 4 which now includes the social edges. The output feature matrix is then calculated for the i -th layer of the GCN, $L^{(i)}$, using Equation. 5 where ρ is an activation function (ReLU), $\tilde{A}$ is the normalized adjacency matrix, and $W^{(i)}$ is a trainable weight matrix for the i -th layer with dimensions $R^{d_{i-1} \times d_i}$.

$L^{(0)} = X$ is the input feature matrix of the GCN while the outputs are treated as the final representation for documents which is fed to a softmax layer for classification. Additionally, the document embeddings are then fed directly to a dense layer with softmax activation. Afterwards, the predictions of both the BERT and GCN components are linearly interpolated to obtain the final classification which is described in Equation. 6. For the BERT model, ‘dost-asti/roBERTa-tl-sentiment-analysis’ (Visperas et al., 2023) is selected over ‘jcbllaise/roberta-tagalog-base’ (Cruz & Cheng, 2021) as it is specifically pre-trained for sentiment analysis on Filipino political sentiment in social media.

For the hyperparameters, a feature dimension of 768 was employed to match the RoBERTa-base encoder output, with a GCN hidden layer of 200 dimensions and a dropout rate of 0.5 for regularization. For optimization, the Adam optimizer is employed with learning rates of 1×10^{-5} for BERT and 1×10^{-3} for GCN components. Training utilized a batch size of 32 over 5 epochs, with early stopping patience of 3 epochs. The interpolation parameter λ was set to 0.7 based on Lin et al. (2021)’s findings. The maximum sequence length for tokenization was set to 128 tokens. Furthermore, a pre-finetuning strategy with 10 epochs at 2×10^{-5} learning rate before joint training.

$$V = V_{\text{doc}} \cup V_{\text{word}} \quad (1)$$

$$E = E_{\text{TF-IDF}} \cup E_{\text{PPMI}} \cup E_{\text{social}} \quad (2)$$

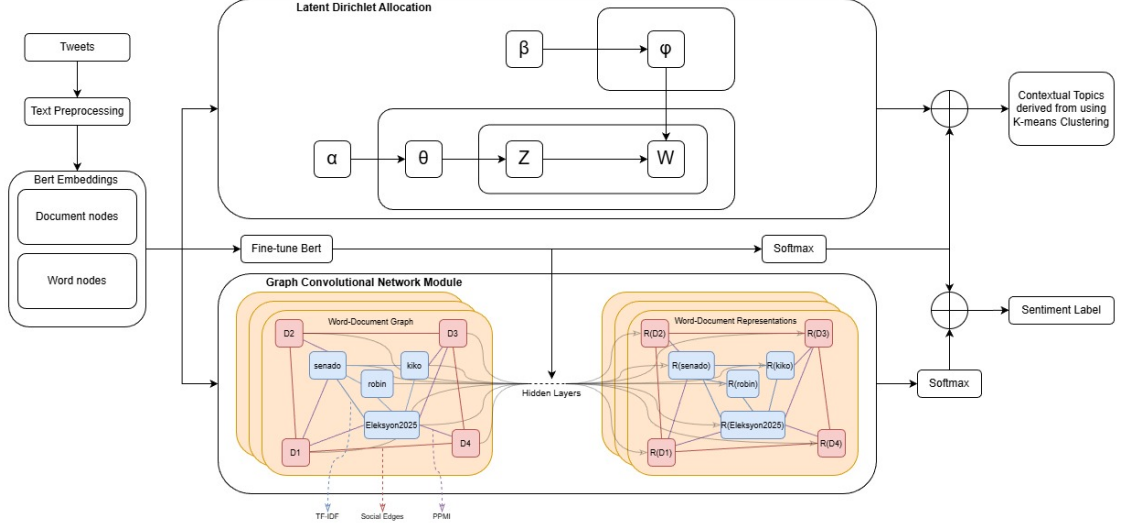


Figure 3. The proposed architecture (extension of EmoBertGCN)

$$X = \begin{pmatrix} X_{\text{doc}} \\ 0 \end{pmatrix}_{(n_{\text{doc}} + n_{\text{word}}) \times d} \quad (3)$$

$$A_{ij} = \begin{cases} \text{PPMI}(i, j), & \text{if } i, j \in V_{\text{word}}, i \neq j \\ \text{TF-IDF}(i, j), & \text{if } i \in V_{\text{doc}}, j \in V_{\text{word}} \\ 1, & \text{if } i, j \in V_{\text{doc}} \text{ and social edge exists} \\ 1, & \text{if } i = j \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

$$L^{(i)} = \rho(\tilde{A}L^{(i-1)}W^{(i)}) \quad (5)$$

$$Z = \lambda Z_{\text{GCN}} + (1 - \lambda)Z_{\text{BERT}} \quad (6)$$

Experimental Validation

Experimental validation followed 10-fold cross-validation with macro F1-score as the primary evaluation metric. Within each fold, the training pool was further split into training (85%) and validation (15%) sets. Conversation threads were preserved across folds to prevent data leakage. Macro-averaged F_1 -score was used as the primary metric, consistent with established practices in sentiment analysis evaluation (Nakov et al., 2019; Rosenthal et al., 2019; Sebastiani, 2015). In determining the significance of the results, a one-tailed t-test was conducted to assess whether our model outperforms the baseline models, which are listed as follows:

- RoBERTa Sentiment (Visperas et al., 2023)
- EmoBertGCN (Aquino et al., 2025)
- Our Proposed Model (EmoBertGCN with Social Edges)

Topic Modeling

The study employed the use of Aquino et al.’s **Emotion-based Topic Modeling using a Hybrid model of BERT and LDA (Latent Dirichlet Allocation)** to assess the sentiments of all the tweets regarding the Philippine general midterm elections. BERT provides the contextual, syntactic, and semantic meaning of the text, while LDA acts as a thematic and topical feature extractor by providing the topic distribution of each document. When combined, it is employed to reveal unseen topics within each sentiment category in the dataset. Given that concatenating BERT embeddings and LDA topic distribution vector results in a high-dimensional vector, it must be reduced in order to better visualize and cluster the topics. Hence, Uniform Manifold Approximation and Projection (UMAP) was used to reduce its dimensions (McInnes et al., 2020). Afterwards, K-means clustering was used to derive the topic clusters based on the output of the BERT-LDA hybrid model. UMass

Coherence Index (UCI) and UMass Coherence Scores was computed to objectively measure the interpretability of the topics derived from the K-means clustering step, as shown in Equation. 7 and Equation. 8, respectively (Röder et al., 2015). Lastly, A log-likelihood ratio Equation. 9, which was adopted from the methodology of Chang et al. (2022), was used to extract sentiment words to better understand the dataset’s intrinsic pattern by creating word clouds.

$$C_{UCI} = \frac{2}{N(N-1)} \sum_{i=1}^{N-1} \sum_{j=i+1}^N \log \frac{P(w_i, w_j) + \epsilon}{P(w_i) \cdot P(w_j)} \quad (7)$$

$$C_{UMass} = \frac{2}{N(N-1)} \sum_{i=1}^{N-1} \sum_{j=i+1}^N \log \frac{P(w_i, w_j) + \epsilon}{P(w_j)} \quad (8)$$

$$LLR(w, T) = 2 \cdot \log \left(\frac{p(w|T)^k (1 - p(w|T))^m p(w|\neg T)^l (1 - p(w|\neg T))^n}{p(w)^{k+l} (1 - p(w))^{m+n}} \right) \quad (9)$$

RESULTS AND DISCUSSION

Model Evaluation

This chapter discusses the results in two main sections. The first section discusses the training and evaluation of the three models: a RoBERTa baseline, the BertGCN baseline, and the extended BertGCN with conversation edges. The models were evaluated using a 10-fold cross validation on a labeled dataset. Despite the intuition that conversation structure could mitigate class imbalance, statistical analysis reveals no significant performance gain from the graph-based models. The second section applies the best-performing model to infer sentiment on 69,335 unlabeled tweets, revealing a dominant negative sentiment class. Topic modeling using BERT-LDA further identifies five interpretable thematic clusters during the Philippine election period.

Table 3 shows the full metrics of all the models evaluated in the experiment. All three models achieved comparable macro F1 scores: RoBERTa baseline (0.8534 ± 0.0293), BertGCN without conversation edges (0.8456 ± 0.0253), and BertGCN with conversation edges (0.8487 ± 0.0254). The relatively high standard deviation values across all metrics indicate that the performance of each model oscillates significantly per fold, suggesting that the performance of all models was sensitive to the composition of the folds.

The performance metrics also indicate that all three models demonstrated strong and consistent performance on the negative and neutral classes, with macro F1 scores of around 0.92 and 0.91, respectively, across all configurations can be seen in Table 3. The consistency of results illustrates that all 3 models were able to effectively learn these two classes. This is further reflected in the overall confusion matrices as seen in Table 4 where the models were able to correctly classify a majority of the negative and neutral samples compared to the positive samples.

In contrast, all models performed considerably worse on the positive class, with

Table 3. Per-class and macro-averaged performance of the three models over 10-fold cross-validation (mean $\pm$ std).

Class	Metric	RoBERTa	BertGCN (no conv)	BertGCN (+ conv)
Positive	Precision	0.7042 ± 0.0772	0.7833 ± 0.0831	0.7545 ± 0.0869
	Recall	0.7317 ± 0.0632	0.6394 ± 0.0742	0.6752 ± 0.0840
	F1	0.7155 ± 0.0586	0.6984 ± 0.0463	0.7050 ± 0.0442
Negative	Precision	0.9237 ± 0.0305	0.9045 ± 0.0371	0.9121 ± 0.0286
	Recall	0.9357 ± 0.0194	0.9484 ± 0.0245	0.9362 ± 0.0337
	F1	0.9294 ± 0.0191	0.9254 ± 0.0213	0.9236 ± 0.0231
Neutral	Precision	0.9334 ± 0.0284	0.9228 ± 0.0375	0.9234 ± 0.0351
	Recall	0.8996 ± 0.0469	0.9065 ± 0.0511	0.9132 ± 0.0428
	F1	0.9152 ± 0.0238	0.9131 ± 0.0249	0.9175 ± 0.0279
Macro	Precision	0.8538 ± 0.0308	0.8702 ± 0.0304	0.8633 ± 0.0315
	Recall	0.8556 ± 0.0317	0.8314 ± 0.0291	0.8415 ± 0.0279
	F1	0.8534 ± 0.0309	0.8456 ± 0.0267	0.8487 ± 0.0268
Accuracy	F1	0.9000 ± 0.0251	0.8980 ± 0.0242	0.8982 ± 0.0268
Weighted	Precision	0.9031 ± 0.0222	0.8986 ± 0.0219	0.8996 ± 0.0221
	Recall	0.9000 ± 0.0251	0.8980 ± 0.0242	0.8982 ± 0.0268
	F1	0.9006 ± 0.0243	0.8957 ± 0.0240	0.8971 ± 0.0253

the macro F1 scores of 0.70 across all models and all folds. As shown in Table 4, the positive class shows the most misclassifications as either the negative or neutral classes, while the reverse (negative and neutral samples being misclassified as positive) occurred significantly less. This asymmetric pattern is consistent with the class imbalance present in the dataset since the model will better learn to classify the class/es with the most documents. Consequently, these results suggest that, since it is consistent across all three models, class imbalance is a more dominant influence on performance than any architectural differences between models.

This is further validated by the results of the one-tailed t-test gathered from the

Table 4. Aggregated confusion matrices over 10-fold cross-validation for all three models.

		RoBERTa (Baseline)			BertGCN (no conv)			BertGCN (+ conv)		
		<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>
Actual	<i>Pos</i>	369	88	46	321	118	64	340	107	56
	<i>Neg</i>	84	2140	62	52	2169	65	76	2139	71
	<i>Neu</i>	76	92	1507	43	114	1518	47	99	1529

Note: Rows represent true labels; columns represent predicted labels.

experiments. Statistical significance was assessed using exact joint permutation tests following the Westfall–Young procedure. Since both comparisons share the BertGCN (+conv, $m = 0.7$) scores as a common group, the standard Bonferroni correction would be over-conservative due to the positive correlation between tests. With $n = 10$ folds, all 184,756 combinations were enumerated exhaustively, yielding exact (non-approximate) marginal and joint p -values. The balance parameter $m = 0.7$ was pre-specified following Lin et al. (2021). All tests are one-tailed at $\alpha = 0.05$.

Table 5. Statistical comparison of macro F1 scores across 10-fold cross-validation.

Comparison	Mean A	Mean B	Cohen’s d	p-value	Sig.
BertGCN (+conv) vs. RoBERTa	0.8487	0.8534	−0.163	0.642	No
BertGCN (+conv) vs. BertGCN (no conv)	0.8487	0.8456	+0.114	0.401	No
Joint FWER (Westfall–Young)				0.647	No

Note: Exact permutation tests with Westfall–Young joint FWER correction. All tests are one-tailed (hypothesis: BertGCN with conv > baseline).

As shown in Table 5, neither comparison showed a significant performance increase. The proposed BertGCN with conversation edges did not significantly outperform the RoBERTa baseline ($p = 0.642$, $d = -0.163$) or the BertGCN baseline ($p = 0.401$, $d = 0.114$). The joint family-wise error rate ($p = 0.647$) further confirms that the family of tests produced no significant result. Lastly, the small effect sizes confirm that the models’ performances are comparable.

The insignificant results may be attributed to the sparsity of conversation-connected posts in the dataset. The addition of conversation edges is premised on the assumption that directly connected posts carry a meaningful relational signal that can inform sentiment classification to be robust against class imbalance issues. However, when the proportion of such connections is low, the graph component has limited opportunity to propagate class-discriminative information across posts, and the added edges may contribute insufficient signal to produce a measurable difference in performance. As a result, the GNN component had little to learn from the edges connecting posts within the same conversation. This provides a structural explanation for the lack of significant performance gains over the baseline BertGCN.

The lack of performance gain of the BertGCN over the RoBERTa baseline contrasts with the results of Aquino et al. (2025)’s experiments, which were under more severe class imbalance conditions. This case is most noticeable with their neutral class, which only comprised 7.3% of samples (583 out of 8,035), which, when compared to the class imbalance issue of this study, is more severe than the dataset used in this study. Furthermore, the BERT component model utilized in Aquino et al. (2025)’s paper was a general-purpose Filipino pretrained language model. On the other hand, this study employed the use of a Filipino pretrained RoBERTa model that is specialized for sentiment classification on code-switched Tagalog text (Visperas et al., 2023). This difference in pretraining provided the models used in this study with stronger sentiment-relevant representations, raising the performance ceiling and further limiting any room for improvement provided by the graph component of BertGCN.

The posited improved towards the positive class is insignificant across any configuration. All three models yielded nearly identical positive-class F1 scores (≈ 0.70), indicating that message-passing did not meaningfully enrich representations of positive-class instances. Two confounding factors prevented a fair test of the underlying hypothesis. First, the specialized RoBERTa backbone (fine-tuned on code-switched Tagalog sentiment data)

already partially compensated for class imbalance through strong lexical and pragmatic priors, raising the performance ceiling and leaving little residual imbalance for the GCN to correct. Second, graph sparsity in the training set severely limited the GCN’s opportunity to propagate class-discriminative signals. Under these conditions, the null result reflects the data set and the backbone constraints rather than a failure of the core hypothesis that the conversation structure can inform sentiment classification under imbalance.

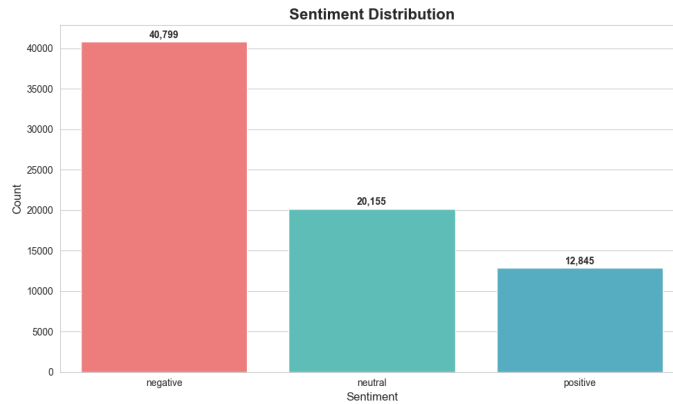


Figure 4. Distribution of sentiment labels in the full dataset. The X-axis displays the three sentiment categories and Y-axis the frequency of sentiment class

Topic Modeling

Given that the proposed BertGCN extension did not improve over the plain RoBERTa baseline, the RoBERTa model was used to conduct inference over the remaining 69,335 unlabeled tweets. Similar to the findings of Aquino et al. (2025), the negative sentiment class has the largest number of tweets as seen in Figure 4, with the neutral and positive classes trailing far behind. The domination of the negative sentiment class in this dataset reflects the findings of Schöne et al. (2021), which highlights the further propagation of the negative sentiment in comparison to positive ones. Furthermore, Figure 5 shows that the average tweet length of negative sentiment labeled tweets is around 157 characters, while tweets under the neutral class have the longest average tweet character length (244 characters).

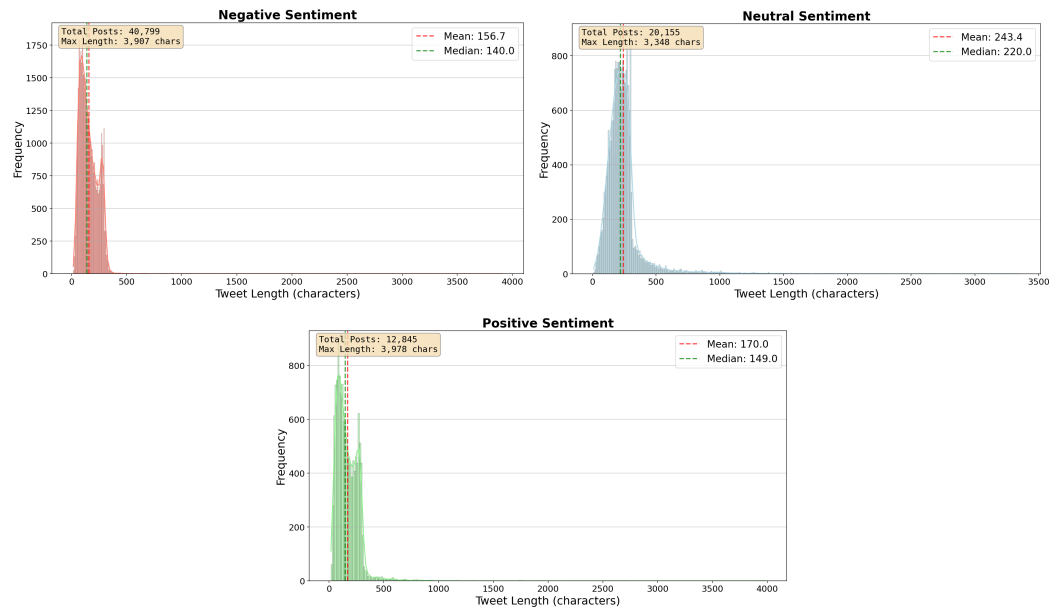


Figure 5. Distribution of tweet length by sentiment class with the X-axis representing tweet length and Y-axis representing frequency of tweets

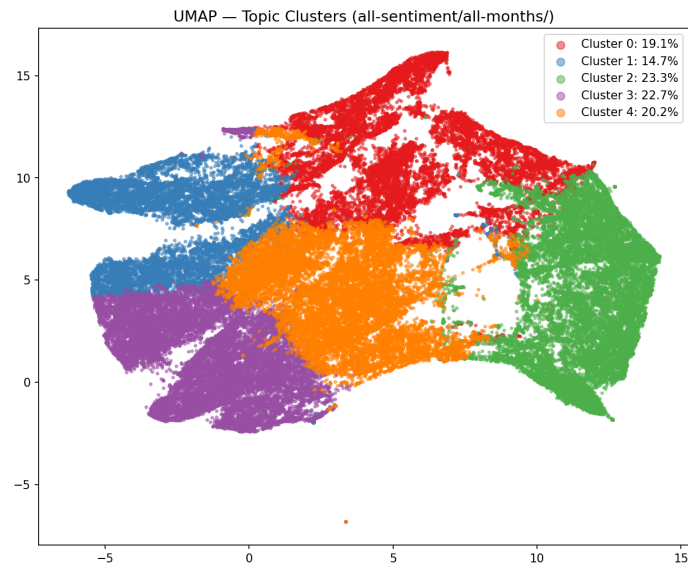


Figure 6. UMAP visualization of topic clusters. Since the dimensions of the BERT-LDA output is more than 2, the X and Y axis does not correspond to any specific features. This visualization just represents the closeness of the tweets within a topic cluster.

Topic coherence was evaluated using UMass and UCI metrics (Chang et al., 2022). Both the UMass and UCI were calculated using the same 73,799 tweets dataset. For UMass, the study used a document-level co-occurrence with boolean document frequencies, while UCI used a sliding window of 10 tokens, justified by the short length of tweets (average < 200 characters). Under this setup, the 5-cluster solution achieved the highest average UCI score (-1.913), outperforming 3 clusters (-2.473) and 4 clusters (-2.486). Although the 5-cluster solution yielded a lower (more negative) UMass score (-3.352) than the 3-cluster solution (-2.985), this study prioritized UCI for topic selection, as it is less sensitive to document frequency biases and more reflective of word association strength. However, the topic modeling results for 3 and 4 clusters were added in the appendix for future reference. The visualization of the 5 cluster output of the Bert-LDA pipeline is shown in Figure 13.

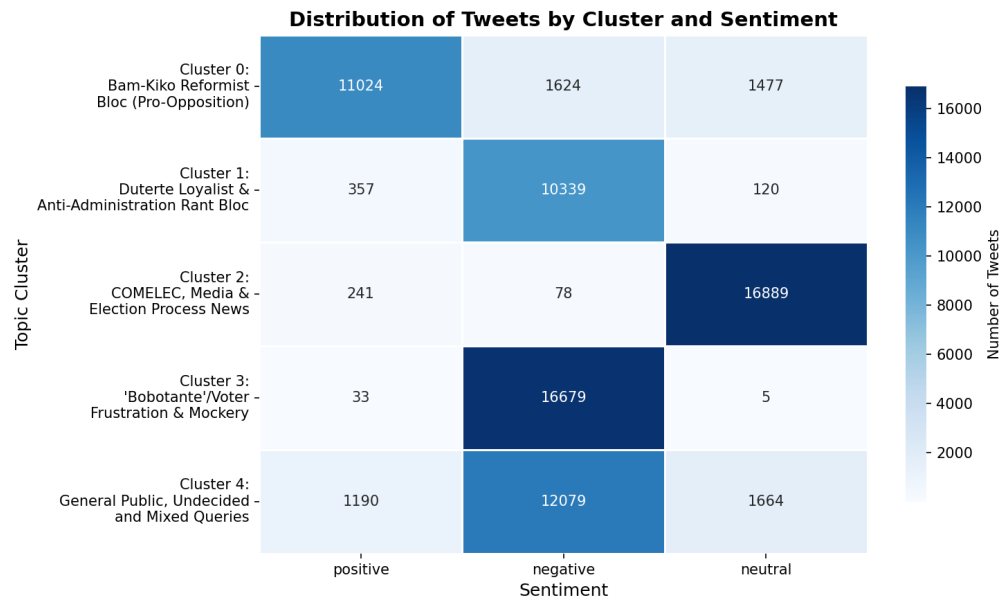


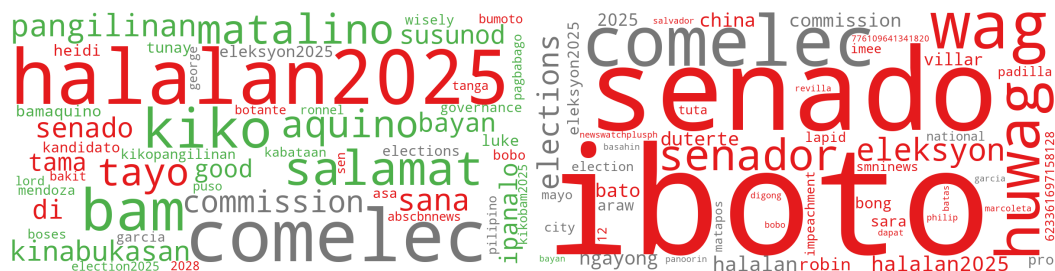
Figure 7. Distribution of tweets per cluster and sentiment class. The X-axis displays the three sentiment categories and Y-axis the cluster categories

Figure 7 illustrates the sentiment class distribution per cluster, with cluster 0 being predominantly positive, cluster 2 negative, and clusters 1, 3, and 4 neutral. It is also determined that despite cluster 1 having the least amount of posts under it, most other

clusters hover around the same number of posts. Figure 12 further clarifies the interpretation of these clusters by introducing the top 50 words present in each cluster. Based on the LLR results, the following labels are derived from the top 100 words present in each cluster.

Cluster 0 is examined to be focused on the Pro-Opposition camp present in the current political sphere of the Philippines. This cluster is heavily dominated by keywords such as *bam*, *kiko*, *pangilinan*, *aquino*, *salamat*, *matalino*, *tama*, *tunay*, good, and governance. However, it does contain some negative labeled words such as *bobo*, *bakit*, and the like. Cluster 1, on the other hand, is titled as the Anti-Administration and Duterte Loyalist Bloc, as it features the keywords Duterte, Sara, Digong, Bato, Robin, Villar, Bong, Imee, PDP, Quiboloy, which are the names of key figures and organizations in that sector, which are mostly found in negative posts. Furthermore, it contains derogatory terms such as *bobo* and *tuta*, which further solidify its theme as a cluster laden with negative sentiment. The same negative dominant sentiment can be seen in Cluster 3, as it features more derogatory terms, which are identified as extremely characteristic of the cluster. Unlike cluster 1, however, this cluster is not endorsing candidates, but rather, is mocking or lamenting the quality of voters. Its theme can be represented as the general frustration of Filipinos with their fellow Filipino voters' intelligence, possibly from a more educated or cynical segment of social media.

On the other hand, cluster 2 tackles the more neutral side of the sentiment spectrum. Here, the top keywords follow along the lines of news and election processes announcements, which are seen in the following keywords: elections, *mayo*, *bilang*, *proklamasyon*, automated, canvassers, disqualification, counting, online, source, *detalye*, *ulat*, *radyopilipinas* (news outlet), *rp1news* (news outlet), *news5* (news outlet). There is no strong candidate endorsement except for low-weight candidate names. The tone is factual and news-oriented. Cluster 4 follows along the neutral dominant sentiment, but this time tackles the curiosity and queries made by the general public. This cluster has a mix of keywords pertaining to the election procedures, such as commission, elections, 2025, *mayo*, city, pnp (Philippine National Police), *senado*. In addition, it also contains curiosity words like *sino*, *pwede*, *kahit*,



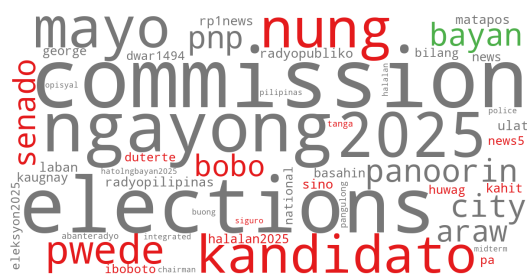
Cluster 0: Bam-Kiko Reformist Bloc
(Pro-Opposition)

Cluster 1: Duterte Loyalist /
Anti-Administration Rant Bloc



Cluster 2: COMELEC & Media / Election
Process News

Cluster 3: "Bobotante" / Voter Frustration &
Mockery



Cluster 4: General Public / Undecided /
Mixed Queries

Figure 8. Word Clouds with Topic and LLR-based sentiment information of clusters. Words are colored based on the dominant sentiment of the documents they appear in most frequently: green for positive, red for negative, and grey for neutral.

siguro, and *iboboto*. It lacks the strong pro-BamKiko signal of Cluster 0 and the negative sentiment signal of Cluster 1. It appears to be undecided, or the general public is asking questions about candidates and processes.

It is also important to note that in clusters 0 and 1, obfuscated Twitter usernames were captured. Additionally, hashtags which were not used during the data collection, such as “#51pangilinankiko”, “#5aquinobam”, “#votewisely”, “#dapattotoo”, indicate that elements unique to the social media environment play an important role in online political discussion, as first posited by Aquino et al. (2025).

SUMMARY AND CONCLUSION

This paper extended the EmoBertGCN framework of Aquino et al. (2025) by incorporating conversation-based social edges into its graph structure and using tweets from the 2025 Philippine general midterm elections as the corpus. Despite the insignificant difference in performance between the proposed integration of the conversational structure present in social media and the BertGCN framework, the integration provided meaningful insight into the limitations of the dataset used to implement it. Specifically, the number of conversation-connected posts was too sparse, constituting only 23.5% of the training set, which limited the conversational context's influence during the model's training. Furthermore, the domain-specialized RoBERTa pretrained specifically for code-switched Tagalog sentiment classification achieved a high metric that beat both the baseline BertGCN and the study's methodology, necessitating substantial improvements by both the baseline BertGCN and this study's methodology to demonstrate a significant contribution.

Notwithstanding the null result, the research still produced several contributions in both the field of NLP and in future analysis on online political discourse in the Philippines. A dataset of both manually and machine-annotated 73,799 tweets situated during a key electoral period has been produced, which serves as a resource for future NLP research on Tagalog and code-mixed text. Additionally, the adopted topic modeling from Aquino et al. (2025)'s paper identified five interpretable discourse clusters: a Pro-Opposition cluster, an Anti-Administration and Duterte Loyalist cluster, an election news and COMELEC cluster, a voter frustration and mockery cluster, and a general public inquiry cluster.

For future research, it is recommended to expand the applications of these models on different social media platforms, while also taking into consideration the interaction structure unique to these platforms. Most importantly, it is advised to future researchers to collect as much conversation information as possible in order to confirm the impact of its integration on the graph structure of the BertGCN framework. The utilization of newer GNN

architectures could be supplied to the framework rather than GCN, in order to account for the heterophily characteristic of political discourse (Chamberlain et al., 2021; Hevopathige et al., 2025; Zhu et al., 2020). Lastly, to ensure more reliable annotation of tweet sentiment, future researchers are recommended to explore more thorough guidance of experts. Together, these directions would enable a more complete evaluation of the theoretical premise that conversational context carries meaningful sentiment information for graph-based learning.

APPENDIX A: PERFORMANCE METRICS PER FOLD

Table 6. Fold 1: Per-class performance metrics for all three models

Class	Metric	RoBERTa	BertGCN (no conv)	BertGCN (+ conv)
Positive	Precision	0.7021	0.8621	0.7692
	Recall	0.7333	0.5556	0.6667
	F1	0.7174	0.6757	0.7143
Negative	Precision	0.9423	0.9161	0.9363
	Recall	0.9496	0.9729	0.9690
	F1	0.9459	0.9436	0.9524
Neutral	Precision	0.9415	0.9200	0.9477
	Recall	0.9200	0.9200	0.9314
	F1	0.9306	0.9200	0.9395
Accuracy	F1	0.9184	0.9142	0.9268
Macro	Precision	0.8620	0.8994	0.8844
	Recall	0.8676	0.8161	0.8557
	F1	0.8647	0.8464	0.8687
Weighted	Precision	0.9194	0.9124	0.9248
	Recall	0.9184	0.9142	0.9268
	F1	0.9188	0.9097	0.9252

Table 7. Fold 2: Per-class performance metrics for all three models

Class	Metric	RoBERTa	BertGCN (no conv)	BertGCN (+ conv)
Positive	Precision	0.7234	0.7632	0.7568
	Recall	0.7391	0.6304	0.6087
	F1	0.7312	0.6905	0.6747
Negative	Precision	0.9204	0.9020	0.8942
	Recall	0.9487	0.9436	0.9538
	F1	0.9343	0.9223	0.9231
Neutral	Precision	0.9655	0.9389	0.9492
	Recall	0.9282	0.9337	0.9282
	F1	0.9465	0.9363	0.9385
Accuracy	F1	0.9171	0.9052	0.9052
Macro	Precision	0.8698	0.8680	0.8667
	Recall	0.8720	0.8359	0.8302
	F1	0.8707	0.8497	0.8454
Weighted	Precision	0.9183	0.9027	0.9028
	Recall	0.9171	0.9052	0.9052
	F1	0.9174	0.9030	0.9026

Table 8. Fold 3: Per-class performance metrics for all three models

Class	Metric	RoBERTa	BertGCN (no conv)	BertGCN (+ conv)
Positive	Precision	0.5921	0.6418	0.5568
	Recall	0.7258	0.6935	0.7903
	F1	0.6522	0.6667	0.6533
Negative	Precision	0.8607	0.8386	0.8777
	Recall	0.9052	0.9181	0.8664
	F1	0.8824	0.8765	0.8720
Neutral	Precision	0.9514	0.9650	0.9524
	Recall	0.8059	0.8118	0.8235
	F1	0.8726	0.8818	0.8833
Accuracy	F1	0.8448	0.8491	0.8405
Macro	Precision	0.8014	0.8151	0.7956
	Recall	0.8123	0.8078	0.8267
	F1	0.8024	0.8083	0.8029
Weighted	Precision	0.8580	0.8586	0.8622
	Recall	0.8448	0.8491	0.8405
	F1	0.8480	0.8504	0.8469

Table 9. Fold 4: Per-class performance metrics for all three models

Class	Metric	RoBERTa	BertGCN (no conv)	BertGCN (+ conv)
Positive	Precision	0.6087	0.8788	0.8649
	Recall	0.7368	0.5088	0.5614
	F1	0.6667	0.6444	0.6809
Negative	Precision	0.9205	0.8808	0.8812
	Recall	0.9283	0.9662	0.9705
	F1	0.9244	0.9215	0.9237
Neutral	Precision	0.9392	0.9080	0.9241
	Recall	0.8580	0.9136	0.9012
	F1	0.8968	0.9108	0.9125
Accuracy	F1	0.8794	0.8904	0.8947
Macro	Precision	0.8228	0.8892	0.8900
	Recall	0.8410	0.7962	0.8110
	F1	0.8293	0.8256	0.8390
Weighted	Precision	0.8882	0.8902	0.8944
	Recall	0.8794	0.8904	0.8947
	F1	0.8824	0.8831	0.8894

Table 10. Fold 5: Per-class performance metrics for all three models

Class	Metric	RoBERTa	BertGCN (no conv)	BertGCN (+ conv)
Positive	Precision	0.8298	0.8293	0.7885
	Recall	0.7358	0.6415	0.7736
	F1	0.7800	0.7234	0.7810
Negative	Precision	0.9493	0.9612	0.9522
	Recall	0.9321	0.8959	0.9005
	F1	0.9406	0.9274	0.9256
Neutral	Precision	0.8896	0.8278	0.8916
	Recall	0.9477	0.9739	0.9673
	F1	0.9177	0.8949	0.9279
Accuracy	F1	0.9133	0.8923	0.9087
Macro	Precision	0.8896	0.8727	0.8774
	Recall	0.8719	0.8371	0.8805
	F1	0.8795	0.8486	0.8781
Weighted	Precision	0.9131	0.8970	0.9101
	Recall	0.9133	0.8923	0.9087
	F1	0.9125	0.8904	0.9085

Table 11. Fold 6: Per-class performance metrics for all three models

Class	Metric	RoBERTa	BertGCN (no conv)	BertGCN (+ conv)
Positive	Precision	0.6744	0.7297	0.7879
	Recall	0.6170	0.5745	0.5532
	F1	0.6444	0.6429	0.6500
Negative	Precision	0.9174	0.8765	0.9000
	Recall	0.9134	0.9524	0.9351
	F1	0.9154	0.9129	0.9172
Neutral	Precision	0.8802	0.9211	0.8922
	Recall	0.9074	0.8642	0.9198
	F1	0.8936	0.8917	0.9058
Accuracy	F1	0.8795	0.8795	0.8886
Macro	Precision	0.8240	0.8424	0.8600
	Recall	0.8126	0.7970	0.8027
	F1	0.8178	0.8158	0.8243
Weighted	Precision	0.8778	0.8772	0.8852
	Recall	0.8795	0.8795	0.8886
	F1	0.8784	0.8762	0.8845

Table 12. Fold 7: Per-class performance metrics for all three models

Class	Metric	RoBERTa	BertGCN (no conv)	BertGCN (+ conv)
Positive	Precision	0.7333	0.7234	0.7391
	Recall	0.6735	0.6939	0.6939
	F1	0.7021	0.7083	0.7158
Negative	Precision	0.9003	0.9152	0.9179
	Recall	0.9704	0.9593	0.9111
	F1	0.9340	0.9367	0.9145
Neutral	Precision	0.9510	0.9195	0.8485
	Recall	0.8500	0.8562	0.8750
	F1	0.8977	0.8867	0.8615
Accuracy	F1	0.8998	0.8977	0.8768
Macro	Precision	0.8616	0.8527	0.8352
	Recall	0.8313	0.8365	0.8267
	F1	0.8446	0.8439	0.8306
Weighted	Precision	0.9002	0.8970	0.8764
	Recall	0.8998	0.8977	0.8768
	F1	0.8982	0.8967	0.8765

Table 13. Fold 8: Per-class performance metrics for all three models

Class	Metric	RoBERTa	BertGCN (no conv)	BertGCN (+ conv)
Positive	Precision	0.7759	0.8837	0.8409
	Recall	0.8654	0.7308	0.7115
	F1	0.8182	0.8000	0.7708
Negative	Precision	0.9760	0.9464	0.9505
	Recall	0.9227	0.9636	0.9591
	F1	0.9486	0.9550	0.9548
Neutral	Precision	0.9181	0.9412	0.9357
	Recall	0.9515	0.9697	0.9697
	F1	0.9345	0.9552	0.9524
Accuracy	F1	0.9268	0.9382	0.9336
Macro	Precision	0.8900	0.9238	0.9090
	Recall	0.9132	0.8880	0.8801
	F1	0.9004	0.9034	0.8927
Weighted	Precision	0.9303	0.9370	0.9318
	Recall	0.9268	0.9382	0.9336
	F1	0.9278	0.9366	0.9320

Table 14. Fold 9: Per-class performance metrics for all three models

Class	Metric	RoBERTa	BertGCN (no conv)	BertGCN (+ conv)
Positive	Precision	0.7679	0.8182	0.7660
	Recall	0.7679	0.6429	0.6429
	F1	0.7679	0.7200	0.6990
Negative	Precision	0.9188	0.8768	0.8835
	Recall	0.9476	0.9686	0.9529
	F1	0.9330	0.9204	0.9169
Neutral	Precision	0.9532	0.9349	0.9357
	Recall	0.9209	0.8927	0.9040
	F1	0.9368	0.9133	0.9195
Accuracy	F1	0.9127	0.8939	0.8915
Macro	Precision	0.8800	0.8766	0.8617
	Recall	0.8788	0.8347	0.8332
	F1	0.8792	0.8512	0.8451
Weighted	Precision	0.9132	0.8933	0.8898
	Recall	0.9127	0.8939	0.8915
	F1	0.9128	0.8910	0.8892

Table 15. Fold 10: Per-class performance metrics for all three models

Class	Metric	RoBERTa	BertGCN (no conv)	BertGCN (+ conv)
Positive	Precision	0.6341	0.7027	0.6750
	Recall	0.7222	0.7222	0.7500
	F1	0.6753	0.7123	0.7105
Negative	Precision	0.9313	0.9316	0.9277
	Recall	0.9394	0.9437	0.9437
	F1	0.9353	0.9376	0.9356
Neutral	Precision	0.9448	0.9518	0.9568
	Recall	0.9059	0.9294	0.9118
	F1	0.9249	0.9405	0.9337
Accuracy	F1	0.9085	0.9199	0.9153
Macro	Precision	0.8368	0.8620	0.8531
	Recall	0.8558	0.8651	0.8685
	F1	0.8452	0.8635	0.8600
Weighted	Precision	0.9121	0.9206	0.9182
	Recall	0.9085	0.9199	0.9153
	F1	0.9099	0.9202	0.9163

APPENDIX B: CONFUSION MATRICES PER FOLD

Table 16. Fold 1: Confusion matrices for all three models.

		RoBERTa (Baseline)			BertGCN (no conv)			BertGCN (+ conv)		
		<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>
Actual	<i>Pos</i>	33	7	5	25	12	8	30	8	7
	<i>Neg</i>	8	245	5	1	251	6	6	250	2
	<i>Neu</i>	6	8	161	3	11	161	3	9	163

Note: Rows represent true labels; columns represent predicted labels.

Table 17. Fold 2: Confusion matrices for all three models.

		RoBERTa (Baseline)			BertGCN (no conv)			BertGCN (+ conv)		
		<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>
Actual	<i>Pos</i>	34	9	3	29	13	4	28	14	4
	<i>Neg</i>	7	185	3	4	184	7	4	186	5
	<i>Neu</i>	6	7	168	5	7	169	5	8	168

Note: Rows represent true labels; columns represent predicted labels.

Table 18. Fold 3: Confusion matrices for all three models.

		RoBERTa (Baseline)			BertGCN (no conv)			BertGCN (+ conv)		
		<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>
Actual	<i>Pos</i>	45	14	3	43	17	2	49	10	3
	<i>Neg</i>	18	210	4	16	213	3	27	201	4
	<i>Neu</i>	13	20	137	8	24	138	12	18	140

Note: Rows represent true labels; columns represent predicted labels.

Table 19. Fold 4: Confusion matrices for all three models.

		RoBERTa (Baseline)			BertGCN (no conv)			BertGCN (+ conv)		
		<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>
Actual	<i>Pos</i>	42	11	4	29	20	8	32	19	6
	<i>Neg</i>	12	220	5	1	229	7	1	230	6
	<i>Neu</i>	15	8	139	3	11	148	4	12	146

Note: Rows represent true labels; columns represent predicted labels.

Table 20. Fold 5: Confusion matrices for all three models.

		RoBERTa (Baseline)			BertGCN (no conv)			BertGCN (+ conv)		
		<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>
Actual	<i>Pos</i>	39	8	6	34	6	13	41	7	5
	<i>Neg</i>	3	206	12	5	198	18	9	199	13
	<i>Neu</i>	5	3	145	2	2	149	2	3	148

Note: Rows represent true labels; columns represent predicted labels.

Table 21. Fold 6: Confusion matrices for all three models.

		RoBERTa (Baseline)			BertGCN (no conv)			BertGCN (+ conv)		
		<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>
Actual	<i>Pos</i>	29	10	8	27	16	4	26	15	6
	<i>Neg</i>	8	211	12	3	220	8	3	216	12
	<i>Neu</i>	6	9	147	7	15	140	4	9	149

Note: Rows represent true labels; columns represent predicted labels.

Table 22. Fold 7: Confusion matrices for all three models.

		RoBERTa (Baseline)			BertGCN (no conv)			BertGCN (+ conv)		
		<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>
Actual	<i>Pos</i>	33	12	4	34	8	7	34	8	7
	<i>Neg</i>	5	262	3	6	259	5	6	246	18
	<i>Neu</i>	7	17	136	7	16	137	6	14	140

Note: Rows represent true labels; columns represent predicted labels.

Table 23. Fold 8: Confusion matrices for all three models.

		RoBERTa (Baseline)			BertGCN (no conv)			BertGCN (+ conv)		
		<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>
Actual	<i>Pos</i>	45	2	5	38	8	6	37	8	7
	<i>Neg</i>	8	203	9	4	212	4	5	211	4
	<i>Neu</i>	5	3	157	1	4	160	2	3	160

Note: Rows represent true labels; columns represent predicted labels.

Table 24. Fold 9: Confusion matrices for all three models.

		RoBERTa (Baseline)			BertGCN (no conv)			BertGCN (+ conv)		
		<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>
Actual	<i>Pos</i>	43	7	6	36	11	9	36	12	8
	<i>Neg</i>	8	181	2	4	185	2	6	182	3
	<i>Neu</i>	5	9	163	4	15	158	5	12	160

Note: Rows represent true labels; columns represent predicted labels.

Table 25. Fold 10: Confusion matrices for all three models.

		RoBERTa (Baseline)			BertGCN (no conv)			BertGCN (+ conv)		
		<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>	<i>Pos</i>	<i>Neg</i>	<i>Neu</i>
Actual	<i>Pos</i>	26	8	2	26	7	3	27	6	3
	<i>Neg</i>	7	217	7	8	218	5	9	218	4
	<i>Neu</i>	8	8	154	3	9	158	4	11	155

Note: Rows represent true labels; columns represent predicted labels.

[illegible]

Figure 9. 3 Cluster Word Clouds with Topic and LLR-based sentiment information of clusters.

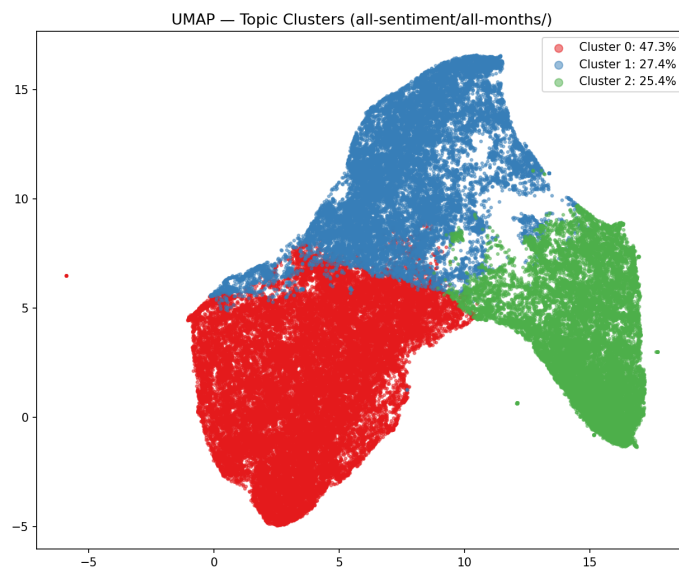


Figure 10. UMAP visualization of 3 topic clusters.

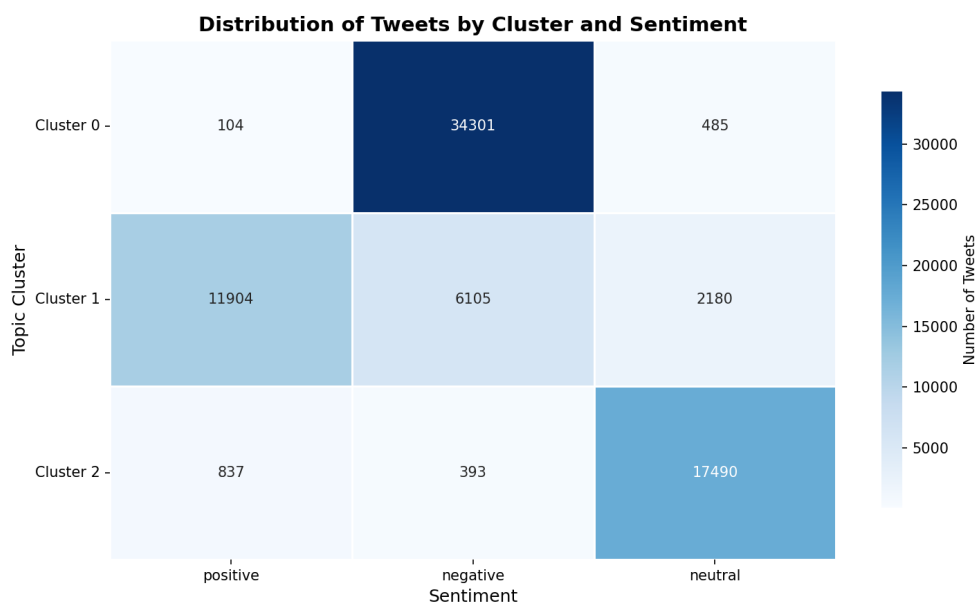


Figure 11. Distribution of tweets per cluster and sentiment class for 3 clusters.

Figure 12. 4 Cluster Word Clouds with Topic and LLR-based sentiment information of clusters.

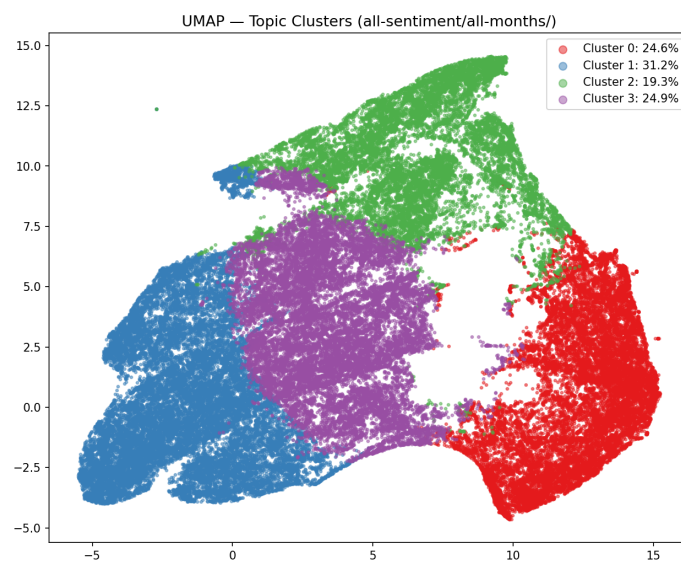


Figure 13. UMAP visualization of 4 topic clusters.

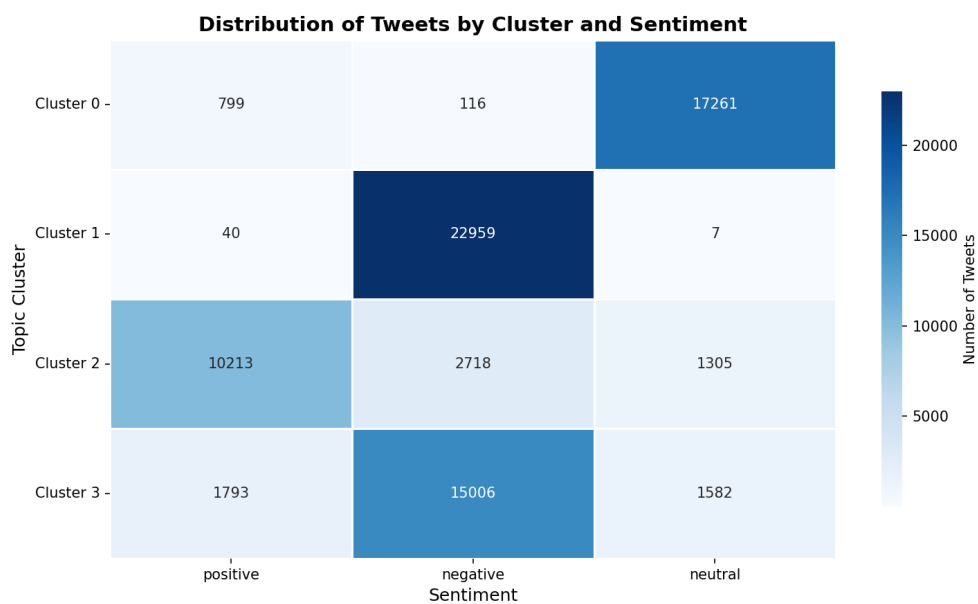


Figure 14. Distribution of tweets per cluster and sentiment class for 4 clusters.

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